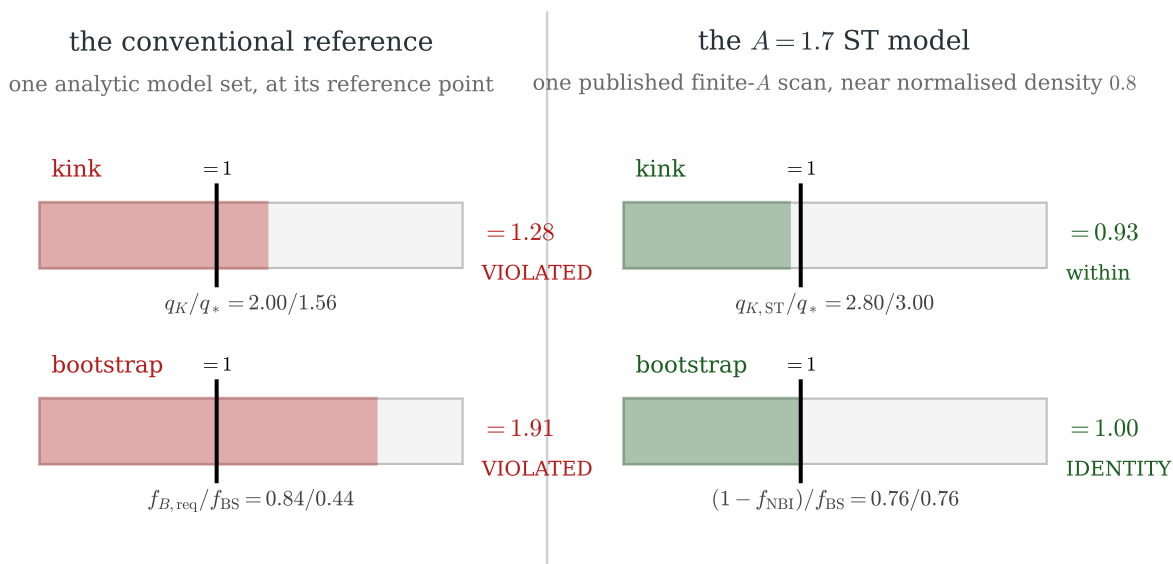


FRIEDBERG'S TWO OBSTRUCTIONS — one model set each side, read the same way

every cell is written DEMAND / ALLOWANCE, so > 1 means violated in all four



WHAT THIS DOES SAY. In the stated finite- A model the kink obstruction closes on its own terms: $q_* \geq 3$ against ST-specific $q_{K,ST} = 2.8$ gives 0.93, and that IS a test that could have failed. It is a model-backed existence result — one configuration in which the number that failed on the left does not fail.

READ THE BOTTOM-RIGHT CELL CAREFULLY. Its 1.00 is an IDENTITY, not a measurement. The scenarios are fully non-inductive — total non-inductive current is computed as bootstrap + NBI — so $f_{BS} + f_{NBI} = 1$ at every point of the scan (60+40 and 84+16 at its two ends). $(1 - f_{NBI})/f_{BS}$ is therefore exactly one everywhere and could not have come out otherwise. The real question the right column raises is the PRICE of that closure: 80 MW of 0.5 MeV negative-ion NBI supplying 0.24 of a 10.8–12 MA plasma. Margin lives there, not in the ratio. And nothing here says every ST is stable or that this one is feasible: the centre-stack question is untouched.

Explanatory sketch; a comparison of declared model results. No magnet, centre stack, HTS or reactor hardware appears: this figure answers the plasma question only.